

KASUN NILAWEERA

Software Engineer — AI Engineering | Data Engineering | Full Stack Development

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PROFILE

Computer Science undergraduate and Software Engineer specializing in AI Engineering, Data Engineering, and Full Stack Development. Experienced in building intelligent applications, scalable data platforms, and production-ready AI systems, with hands-on work spanning multi-agent AI systems, serverless cloud architecture, and ETL/data warehousing. Passionate about AI agents, LLM applications, and cloud-native backend engineering.

WORK EXPERIENCE

Software Engineer Intern — Hasthiya IT

July 2025 – July 2026

Contributed to software development projects as part of the engineering team, gaining hands-on industry experience across the full development lifecycle.

PROJECTS

CloudWave Events — AWS Serverless Event Management Platform

Aug 2026

<https://d1glqpr0nzlm26.cloudfront.net>

Built a full-stack, cloud-native serverless event management platform on AWS using Infrastructure as Code. Enabled secure JWT-based authentication (Amazon Cognito), event CRUD and registration via Lambda & API Gateway, secure file uploads with S3 pre-signed URLs, automated email notifications (SES), scheduled reminder workflows (EventBridge), global content delivery (CloudFront), and CI/CD automation with GitHub Actions, with centralized monitoring via CloudWatch.

Tech Stack: *React.js, Node.js, AWS Lambda, API Gateway, Cognito, Aurora Serverless PostgreSQL, S3, CloudFront, SES, EventBridge, CloudWatch, GitHub Actions, Serverless Framework*

Multi-Agent AI Research Assistant

Jul 2026

<https://github.com/kasunwathsala/Multi-agent-AI-researcher>

Built a multi-agent AI system automating the research-to-report workflow, from clarifying an ambiguous query to delivering a structured, sourced report. Orchestrated four specialized agents (Planner, Research, Analyst, Writer) with LangGraph; the Planner uses LangGraph's human-in-the-loop interrupt/resume pattern to pause and clarify scope mid-execution. Agents communicate via Pydantic-based structured outputs for reliable, type-safe handoffs. Exposed via a FastAPI backend with thread-based session management, connected to a custom Next.js + TypeScript chat frontend.

Tech Stack: *LangChain, LangGraph, FastAPI, Pydantic, Next.js, TypeScript, Tailwind CSS, OpenAI*

ABC HUB ETL Pipeline

Jun 2026 – Jul 2026

Built an end-to-end ETL pipeline with Apache NiFi and PostgreSQL to automate data extraction, transformation, validation, and loading into a Star Schema data warehouse. Implemented workflow orchestration, error handling, and logging for a reliable, scalable analytical reporting pipeline. Strengthened practical skills in ETL processes, SQL, data warehousing, and dimensional modeling.

Tech Stack: *Apache NiFi, PostgreSQL, ETL, Data Warehousing, Star Schema, Workflow Orchestration*

Multi-Agent Travel Planner

2026

<https://github.com/kasunwathsala/MULTI-AGENT-TRAVEL-PLANNER>

Developed an AI-powered travel planning platform generating personalized itineraries via multi-agent workflows: a Requirement Agent for understanding preferences, a Planner Agent for optimized itineraries, and a Booker Agent for booking recommendations. Implemented agent collaboration with LangChain and LangGraph using OpenAI models, with a FastAPI backend for scalable AI-driven travel planning.

Tech Stack: *LangChain, LangGraph, OpenAI, FastAPI, Python*

SunLeaf Energy — Solar Monitoring Dashboard

Jan 2026 – Feb 2026

<https://fed-4-front-end-wathsala.netlify.app/>

Developed a full-stack Solar Dashboard for real-time solar energy monitoring, anomaly detection, and billing automation. Built interactive dashboards with live energy analytics, historical reporting, and weather data integration, plus secure authentication, Stripe payment processing, automated background jobs, and scalable REST APIs.

Tech Stack: *React, Node.js, MongoDB, TypeScript, Redux, Stripe, Zod, Tailwind CSS, shadcn/ui*

Salon Booking System

May 2025

<https://saloon-booking-system.vercel.app/>

Built a responsive salon booking website with dynamic service listings, stylist profiles, a multi-step appointment booking workflow, customer testimonials, and contact management, using component-based architecture and responsive design.

Tech Stack: *Next.js, React, TypeScript*

EDUCATION

BSc (Hons) in Computer Science Sep 2023 – Present

University of Westminster (at Informatics Institute of Technology, Sri Lanka)

Foundation Certificate in Higher Education — Graduated with Distinction Jan 2023 – Aug 2023

Informatics Institute of Technology

TECHNICAL SKILLS

Languages & Frameworks	Python, TypeScript, JavaScript, Java, React, Next.js, Node.js, Express, Angular, FastAPI, Spring Boot
AI / ML & Agentic Systems	OpenAI, LangChain, LangGraph, AutoGen, HuggingFace, PyTorch, Pinecone, LlamaIndex
Cloud & DevOps	AWS (Lambda, API Gateway, Cognito, S3, SES, EventBridge, CloudFront, CloudWatch), Google Cloud Platform, Azure ADLS, Docker, Kubernetes, GitHub Actions (CI/CD)
Data Engineering	Apache Spark, Apache Hadoop, Apache NiFi, ETL Pipelines, Data Warehousing, Star Schema, Dimensional Modeling
Databases & Backend	PostgreSQL, Amazon Aurora Serverless, Firebase, Prisma
Other Tools	Tailwind CSS, shadcn/ui, Redux, Stripe, Zod